DETECTION AND INTERVENTION SIMULATION FOR POSTPARTUM DEPRESSION RISK

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Panopto video: ***https://utexas.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=92aca50d-56ac-4e31-adc5-b33400381f5d***

ABSTRACT

Postpartum depression (PPD) is a deeply impactful maternal mental health condition that can silently progress into severe risk if left undetected. Early identification is critical, yet traditional screening often misses cases, especially those expressed in informal, non-clinical contexts. In this high-risk project, I present an experimental framework that integrates emotion trajectory modeling, rule-based risk assessment, and causal inference to analyze authentic but anonymized postpartum narratives sourced from online communities. This pipeline maps sentence-by-sentence emotional flows, classifies posts into distinct emotional arcs, detects high-risk patterns using interpretable keyword and sentiment rules, and simulates hopeful narrative interventions. Using causal forest models, I estimate the heterogeneous impact of positive emotional shifts on reducing risk probability. While my approach is intentionally exploratory and embraces potential points of failure, findings suggest that natural language signals, when examined through dynamic emotional patterns, can provide an early, scalable lens for detecting and mitigating PPD risk.

1. INTRODUCTION

Postpartum depression (PPD) affects 1 in 7 individuals during or after pregnancy, posing serious risks to both maternal and infant well-being. Many cases go undiagnosed due to stigma and difficulties in expressing emotional distress. Traditional screenings often miss early warning signs, especially those embedded in real-time, unstructured narratives shared online.

Digital platforms like Reddit offer spontaneous, emotionally rich accounts of postpartum experiences. These posts provide a unique opportunity for early detection using computational tools that go beyond simple sentiment analysis. Rather than classifying entire posts statically, we model sentence-level **emotion trajectories**, capturing the rise or fall in affective tone that may signal either recovery or escalating risk.

This project uses ***NRCLex*** to extract emotions and build interpretable arcs from Reddit narratives. I then apply a **rule-based model** to classify high-risk posts and simulate text-based **cognitive reframing** as an intervention. Finally, I use **causal forest models** to explore whether **positive emotional shifts** lead to reduced clinical risk.

This exploratory work tackles the challenge of small, noisy, informal data, where emotion-laden language can be sarcastic, metaphorical, or culturally nuanced. While lexicon and causal models can be fragile under such conditions, the insights gained are promising.

By modeling emotion trajectories and linking them to risk, this work lays the foundation for **early-warning systems** that surface signs of distress in digital spaces, before individuals ever reach a clinical setting.

2. RELATED WORK

Several studies have approached postpartum depression detection and emotional trajectory modeling through text analysis:

- **De Choudhury et al.** (2014) [1] analyze user behavior and linguistic signals across pre- and postnatal periods to predict postpartum risk, emphasizing the power of longitudinal data but relying on traditional linguistic features
- **Amir et al. (2017)** [2] (in collaboration with Glen Coppersmith et al.) develop neural user embeddings from social media timelines to predict depression and PTSD status, moving beyond lexicon-based modeling.

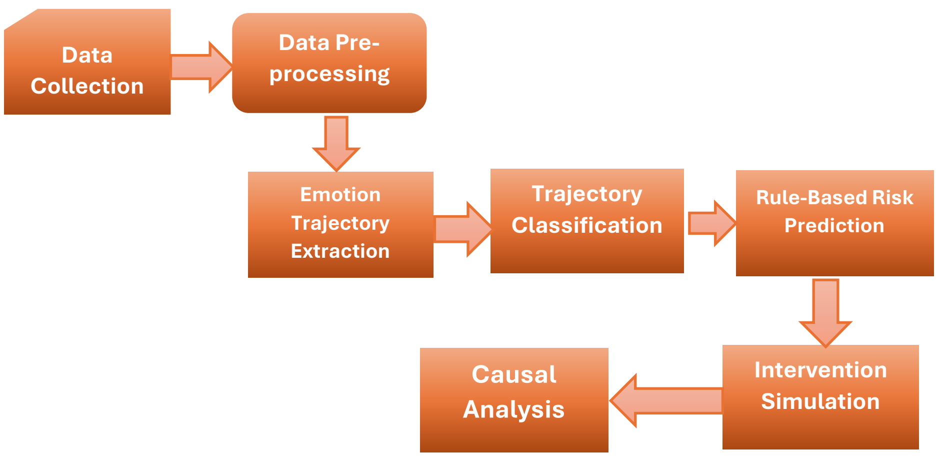
- **Chancellor & De Choudhury (2020)** [3] provide a systematic overview of computational techniques for mental health prediction via social media, comparing lexicon-based and learned models and addressing standards in evaluation

This approach differs by focusing **specifically on postpartum forum narratives**, modeling sentence-level emotion trajectories, integrating *causal inference*, and simulating text-based interventions for actionable early-warning support.

3. METHODOLOGY

The project follows a structured pipeline that transforms raw postpartum narratives into actionable risk insights. Each stage, from text preprocessing to causal effect estimation, builds upon the previous, enabling both descriptive and predictive analysis of emotional trajectories.

Fig. 1: Methodology Workflow



3.1 Data Collection

This study uses a structured CSV dataset of postpartum narratives sourced from online forums like Reddit, where individuals anonymously share personal accounts of emotional, psychological, and physical challenges after childbirth. These free-text posts often reveal distress, daily struggles, and occasionally suicidal ideation, providing rich, underutilized insights into maternal mental health. The language varies widely, reflecting informal discourse, slang, acronyms, and community-specific jargon, offering both analytical challenges and opportunities. Each post was manually annotated with:

- **Trajectory:** The emotional arc across sentences:
  - neg_to_pos: starts negative, ends positive
  - pos_to_neg: starts positive, ends negative
  - stable_neg: remains negative
  - stable_pos: remains positive
- **Risk Label:** Categorical indicator of postpartum depression risk
  - 0: low risk
  - 1: moderate risk
  - 2: high risk

3.2 Preprocessing

The raw Reddit posts are first segmented into individual sentences using sentence tokenization. Common words that add little semantic value (stopwords) are removed using *NLTK's stopword* list, ensuring cleaner input for emotion detection. This step also handles informal language, punctuation inconsistencies, and common acronyms present in user-generated text.

3.3 Emotion Trajectory Extraction

To capture the evolution of emotional tone, I implemented a sentence-level emotion classification pipeline using **NRCLex**, which maps words to eight core emotion categories from the NRC Emotion Lexicon: *joy, sadness, anger, fear, trust, anticipation, disgust,* and *surprise*, along with polarity (positive/negative). For each sentence, I Tokenized and scored words using NRCLex, Identified the dominant emotion by selecting the category with the highest score, Constructed a trajectory as a sequence of dominant emotions in sentence order Transformed the emotion sequence into **valence scores** (e.g., joy = +2, sadness = -2) to enable visualization and quantitative analysis, This process resulted in an emotional timeline for each post, revealing whether emotional tone improved, declined, or remained stable.

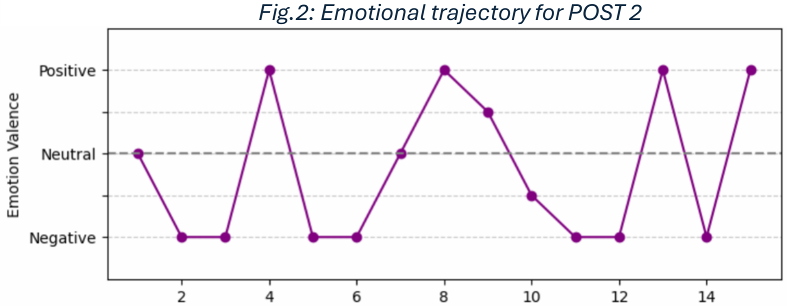
3.4 Trajectory Classification

Using the valence-based emotional sequence, each post were automatically classified into one of four predefined trajectory types: *neg_to_pos, pos_to_neg*, *stable_neg* and *stable_pos.* To evaluate the model's performance, I compared automated labels against manual annotations using a classification report, reporting agreement rate, precision, and recall. I also introduced a **turning point detection** mechanism to identify moments of significant emotional change. The point of largest positive shift in valence between consecutive sentences was marked as a potential cognitive reappraisal event and analyzed qualitatively for content patterns.

To qualitatively illustrate the emotional arc modeling process, let's take a look at an example of *Post 2* from the dataset. These narrative expresses body image concerns and physical recovery struggles six months postpartum. Below is the original text and the sequence of detected sentence-level emotions: *6 months postpartum... my doctor even said, "Wow, pregnancy really wrecked your body." ... I feel so insecure that I look so pregnant... I do small workouts... I'm going to get serious about working out... I hope to get back in shape.*

Detected Emotions: ['neutral', 'disgust', 'disgust', 'positive', 'disgust', 'anger', 'negative', 'positive', 'anticipation', 'fear', 'anger', 'disgust', 'joy', 'anger', 'positive'] These sentence-wise valence scores were then plotted to visualize emotional evolution across the post, as shown below.

Fig.2: Emotional trajectory for POST 2



3.5 Rule-Based Risk Prediction

It is an interpretable, rule-based model to assign **clinical risk levels** to each post. Risk was determined by combining:

- **Trajectory type**: stable_neg and pos_to_neg arcs were more likely to be high-risk.
- **Presence of high-risk keywords**: e.g., "hopeless," "suicide," "self-harm," "overdosed."
- Risk levels were assigned as follows:
  - **2** = High risk (negative trajectory + high-risk keywords)
  - **1** = Moderate risk (negative trajectory without keywords)
  - **0** = Low risk (positive or improving trajectories)

3.6 Intervention Simulation

To explore potential digital interventions, I simulate **cognitive reframing** by appending hopeful, future-oriented statements (e.g., "I am learning to cope and see brighter days ahead") to high-risk posts. These modified posts were re-evaluated by the risk model, and any changes in risk score were recorded to assess the effectiveness of this simple linguistic intervention. Example:

"I cry every night and I don't think I'm a good mom"	***Original Risk:*** *2 (High)*
I cry every night and I don't think I'm a good mom. ***But I know I'm trying my best, and I have people who love me. I'll talk to my doctor."***	***New Risk:*** *1 (Moderate)*

Fig.3: Risk Level before and After hopeful intervention

3.7 Causal Analysis

To estimate the effect of positive emotional shifts on postpartum depression risk, I applied the **CausalForestDML** model from the *econml* library. The treatment variable was whether a post followed a *negative-to-positive* (neg_to_pos) emotional trajectory (*treatment* = 1) or any other trajectory (*treatment* = 0). The binary outcome variable indicated whether the post was classified as high-risk by the rule-based risk model.

Covariates included aggregated counts of each NRC emotion category per post, allowing the model to account for baseline emotional composition when estimating treatment effects. CausalForestDML was chosen for its ability to adjust for observed confounders and to estimate both **Average Treatment Effects (ATE)** and **Individualized Treatment Effects (ITE)**, capturing heterogeneity in response to emotional shifts.

To explore variability in treatment effects across trajectory types, I planned to visualize the distribution of ITEs by group (Fig. 4) and assess which emotion categories contributed most to treatment effect heterogeneity via model-derived feature importances (Fig. 5).

4. RESULT

The risk model flagged **high-risk** posts when **stable_neg** arcs co-occurred with crisis keywords. Posts without such keywords but with prolonged negativity were marked **moderate risk**. Risk **reduction** was observed in several posts after hopeful reframing (e.g., Post 0: risk 2→0; Post 1: 2→1). However, posts with explicit suicidal language (e.g., "overdosed") retained high-risk status, demonstrating a safeguard against **false downscoring**. This reflects **CBT principles**, where **cognitive reframing** reduces hopelessness. The intervention simulates how digital tools could support resilience, though **clinical attention** remains essential for severe cases.

Label	Precision	Recall	F1-score	Support
neg_to_pos	0.44	0.40	0.42	10
pos_to_neg	0.00	0.00	0.00	4
stable_neg	0.74	0.65	0.69	26
stable_pos	0.64	0.64	0.64	11

The automated emotion arc classifier achieved **55% accuracy**, aligning **54.9%** with manual labels. It performed best on **stable arcs** (*stable_neg*: F1 0.69, *stable_pos*: F1 0.64) but struggled with **transition arcs** like *neg_to_pos* (F1 0.42) and *pos_to_neg* (F1 0.00), often misclassifying short or emotionally complex posts due to lexicon-based limitations (e.g., sarcasm, guilt, conflicting tones). The relatively low accuracy (55%) stems primarily from the **inherent mismatch between lexicon-based emotion detection and the informal, nuanced language** of postpartum forum posts. NRCLex assigns emotions purely from pre-defined word–emotion mappings, without understanding sarcasm, mixed emotional states, negations, or context shifts.

Turning points, the largest positive emotional shifts have typically occurred mid-post, linked to self-reflection or changes in the baby's condition. For example: *"Self-love is part of motherhood too, and I make sure to give myself grace"* marked a transition in a *neg_to_pos* post. These moments may offer intervention opportunities and reinforce therapeutic reframing.

Finally, **CausalForestDML** estimated an Average Treatment Effect (ATE) of –0.067 (95% CI: [–0.345, 0.210]), indicating a small but uncertain benefit of *neg_to_pos* shifts. Variability was highest in *stable_neg* and *neg_to_pos* groups. Key emotional drivers of effect included sadness, joy, and fear.

Using the **CausalForestDML** model from the *econml* library, I estimated the individualized treatment effect (ITE) of shifting a post's emotional trajectory from negative-to-positive (*neg_to_pos*) versus other trajectories. The treatment variable was trajectory type (*neg_to_pos* = 1, otherwise = 0), the outcome was high-risk classification (binary), and covariates were aggregated NRC emotion counts per post.

The model, which adjusts for potential confounders, produced both average and heterogeneous treatment effect estimates with 95% confidence intervals. The estimated Average Treatment Effect (ATE) was –0.067 (95% CI: [–0.345, 0.210]), suggesting a small and statistically uncertain reduction in high-risk classification when a post shifted from negative to positive tone.

A boxplot of ITEs (Fig. 4) revealed variability across trajectory groups: some, such as *neg_to_pos*, showed a higher median effect, while others exhibited wide dispersion, indicating inconsistent benefits from positive shifts. Feature importance analysis (Fig. 5) highlighted sadness, joy, and fear as the most influential emotional categories in moderating treatment effects.

The CausalForestDML model's feature importance analysis (Fig. 5) identified which emotional signals most strongly influenced the estimated treatment effect of shifting a post's trajectory from negative to positive. Among the NRC emotion categories, **sadness**, **joy**, and **fear** emerged as the top predictors of how much a post's high-risk classification would change after a positive trajectory shift.

These importances reflect the model's internal structure, where features that consistently improve the estimation of treatment effect heterogeneity are assigned greater weight. The prominence of sadness suggests that reductions in negative affect are central to risk reduction, while joy and fear likely capture more nuanced emotional transitions that either amplify or dampen the benefits of positive reframing.

Fig.4: Estimated Treatment Effect by Trajectory

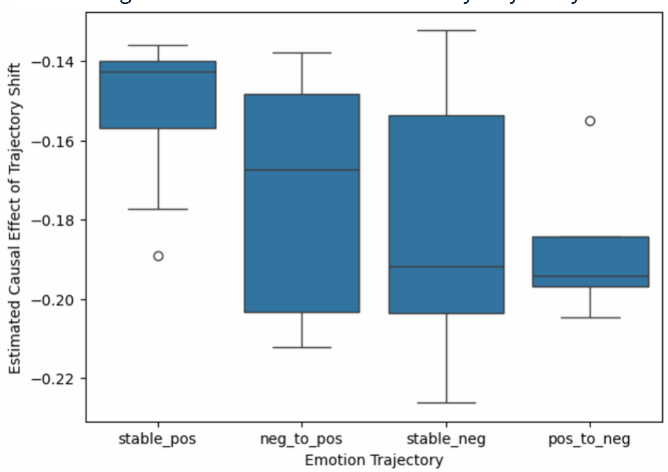
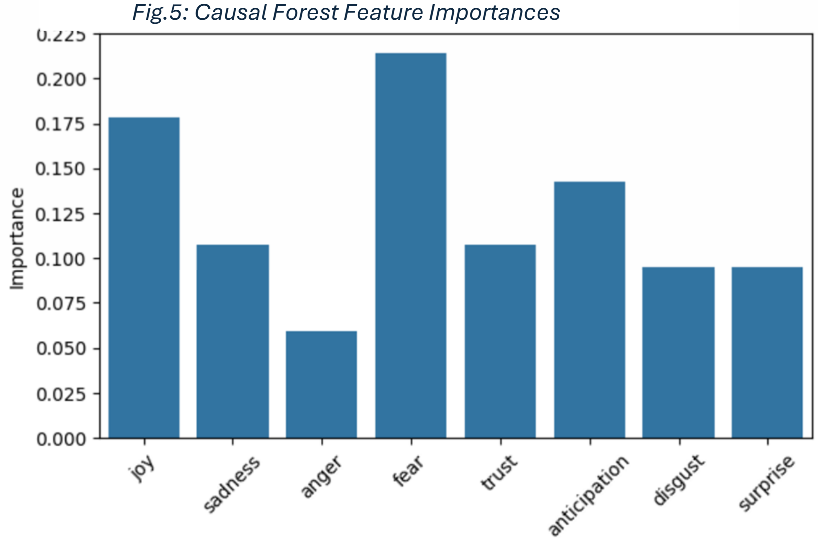


Fig.5: Causal Forest Feature Importances



5. CONCLUSION

This project shows that sentence-level emotion trajectories combined with rule-based risk models can provide an interpretable approach for early detection of postpartum depression risk in online narratives. The system highlights sustained negativity and key distress expressions as signals for potential intervention, and causal analysis offers a way to explore which emotional shifts may influence outcomes.

Beyond classification, the modeling process revealed two actionable insights: certain "turning points" in a narrative may be ideal moments for supportive intervention, and greater emotional diversity in posts could be a resilience indicator worth further study.

Potential applications include:

- Embedding the model into maternal health apps for passive, privacy-conscious screening.
- Supporting moderation in online peer-support communities.
- Providing aggregated early-warning dashboards for clinicians.
- Triggering just-in-time prompts when emotional decline is detected.

By focusing on dynamic emotional patterns rather than static sentiment, this approach could complement clinical care and enable timely, scalable mental health support in digital spaces.

6. LIMITATIONS

- **Lexicon Limitations:** *NRCLex* is simple and interpretable but struggles with informal language, sarcasm, and cultural nuance, leading to misclassification in complex emotional posts.
- **Classifier Performance**: The emotion arc model reached 55% accuracy, with low F1-scores on transitions (e.g., neg_to_pos: 0.42, pos_to_neg: 0.00), highlighting difficulty in capturing sentiment shifts in short or inconsistent texts.
- **Rigid Risk Rules:** The rule-based risk model is transparent but inflexible—hopeful edits didn't reduce risk when self-harm keywords were present, due to strict reliance on fixed rules.
- **Causal Inference Limits:** Causal forest results are affected by small, imbalanced data and unobserved confounders, reducing reliability and interpretability.

7. FUTURE WORK

- Context-Aware Emotion Detection: Replace lexicon-based methods with transformer models (e.g., *ClinicalBERT*, *MentalRoBERTa*) to better handle nuanced emotions, sarcasm, and context.
- Neural Emotion Arc Modeling: Use RNNs, GRUs, or Transformers to learn complex emotion patterns across sentences, improving detection of escalation, volatility, and recovery.
- Improved Risk Prediction: Move beyond rules to a learned model combining arcs, high-risk keywords, and contextual embeddings for a more nuanced, balanced risk score.
- Larger & Clinical Validation: Evaluate on bigger, more diverse datasets with clinical or crowdsourced labels to ensure broader applicability and robustness.
- Ethics & Human Oversight: Integrate human reviewers for flagged content, handle uncertainty, and design ethically with privacy, consent, and responsible intervention in mind.

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